JUSTICE OR PREJUDICE? QUANTIFYING BIASES IN LLM-AS-A-JUDGE

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ABSTRACT

Large language models (LLMs) are increasingly deployed as automated judges to evaluate the quality of model-generated responses—a paradigm known as *LLM-as-a-Judge*. This approach is central to benchmarking, reward modeling, and alignment pipelines, yet its reliability remains undermined by systematic biases and inconsistent judgments. Existing studies examine only a narrow range of bias types, rely on costly human annotation, and lack scalable, reproducible assessment frameworks. In this paper, we introduce CALM (**C**omprehensive **A**ssessment for **L**LM Biases in Judging), an automated benchmark that frames bias evaluation as *attack-and-detect*: principled, semantics-preserving perturbations are applied to inputs, and resulting decision shifts are quantified with unified metrics. CALM covers 12 distinct bias types across three curated datasets (1,439 items total) and evaluates six state-of-the-art judge LLMs—ChatGPT, GPT-4-Turbo, GPT-4o, Claude-3.5, GLM-4, and Qwen2-72B-Instruct. Our experiments reveal that biases persist even in the strongest models: sentiment bias (robustness rate as low as 0.651) and position bias (robustness rate below 0.5 for three or more candidates) are particularly severe, while self-enhancement error rates reach up to 16.1 in certain models. CALM provides a reproducible, automated platform for exposing and quantifying judge biases, offering actionable guidance for building more reliable LLM evaluation pipelines.

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1 INTRODUCTION

The rapid proliferation of large language models (LLMs) has created an urgent need for scalable evaluation methods. Human annotation—long considered the gold standard—is expensive, time-consuming, and difficult to standardize at scale (Ouyang et al., 2022). This has motivated the *LLM-as-a-Judge* paradigm, where powerful LLMs such as GPT-4 serve as automated evaluators, assessing the quality of other models' responses in pairwise comparisons or absolute scoring tasks (Zheng et al., 2024). This paradigm has become integral to open-ended benchmarking (e.g., MT-Bench, Chatbot Arena), reward model training for RLHF, and preference dataset construction (Bai et al., 2022; Rafailov et al., 2024).

Despite its convenience, the reliability of LLM-as-a-Judge has come under increasing scrutiny. Several studies have identified specific failure modes: position bias, where judges favor responses appearing first in the context (Wang et al., 2023; Shi et al., 2024); verbosity bias, where longer responses receive unwarranted preference (Wu & Aji, 2023); and self-enhancement bias, where a model preferentially scores its own outputs (Panickssery et al., 2024; Li et al., 2023). Additional work has highlighted cognitive biases (Koo et al., 2023), authority bias (Chen et al., 2024), and inconsistency under prompt perturbations (Stureborg et al., 2024).

However, existing studies typically examine only one or two bias types in isolation, rely on costly human annotation or hand-crafted perturbations that limit reproducibility, and lack a unified metric framework for fair cross-bias comparison.

To address these gaps, we propose CALM, a **C**omprehensive **A**ssessment framework for **L**LM Biases in Judging. CALM operationalizes bias evaluation as an *attack-and-detect* procedure: for each bias type, we define a principled perturbation function $g(\cdot)$ that modifies either the judge's input (answers or instructions) in a semantics-preserving manner, then measure the resulting shift in judge decisions. This formulation allows us to unify diverse bias types under a common experimental protocol while ensuring that all perturbations can be applied automatically at scale.

CALM makes the following core contributions:

- ▷ **A taxonomy of 12 bias types** affecting LLM-as-a-Judge, spanning answer-related biases (position, verbosity, compassion-fade, fallacy-oversight, sentiment, authority), instruction-related biases (bandwagon, distraction, diversity, chain-of-thought), and scoring-specific biases (self-enhancement, refinement-aware).
- ▷ **An automated, reproducible benchmark** with three curated datasets totaling 1,439 items: a fact-related set (500 items from GSM8K, MATH, and ScienceQA), a refinement-aware set (500 items from CommonsenseQA, Quora-QuAD, and TruthfulQA), and an alignment set (439 pairs derived from five DPO datasets).
- ▷ **Unified evaluation metrics** including Robustness Rate (RR), Consistency Rate (CR), Chain-of-Thought accuracy (Acc_{CoT}), self-enhancement error rate ($\text{ErrorRate}_{\text{SE}}$), and refinement-aware error rate ($\text{ErrorRate}_{\text{RA}}$), enabling fair comparison across all bias types.
- ▷ **A comprehensive evaluation** of six prominent judge LLMs revealing systematic biases even in the most capable models, with actionable guidance on vulnerabilities and mitigations.

Our results confirm that no current judge LLM is bias-free: sentiment bias is the most pervasive vulnerability, position bias becomes catastrophic with more candidates, and self-enhancement varies dramatically across models.

2 RELATED WORK

LLM-as-a-Judge. Zheng et al. (2024) introduced LLM-as-a-Judge with MT-Bench and Chatbot Arena, showing GPT-4 achieves over 80% agreement with human preferences. Subsequent work explored multi-agent evaluation (Li et al., 2023), meta-evaluation (Zeng et al., 2024), and trustworthiness (Sun et al., 2024).

Position and verbosity biases. Wang et al. (2023) showed that swapping response order can flip GPT-4's verdict in a majority of cases. Wu & Aji (2023) found that LLMs favor stylistically polished but factually weaker responses. Shi et al. (2024) systematically investigated position bias.

Cognitive and social biases. Koo et al. (2023) benchmarked six cognitive biases in LLM evaluators. Chen et al. (2024) studied authority, gender, and misinformation biases, showing cutting-edge models remain vulnerable.

Robustness and consistency. Stureborg et al. (2024) showed that LLMs exhibit familiarity bias and anchoring effects, and are inconsistent across rephrased prompts.

Self-enhancement. Panickssery et al. (2024) discovered that LLMs such as GPT-4 can recognize and prefer their own outputs, establishing a linear correlation between self-recognition and self-preference.

In contrast to prior work, CALM unifies 12 bias types under a single automated framework with shared metrics, enabling systematic cross-bias and cross-model comparison.

3 THE CALM FRAMEWORK

3.1 PROBLEM FORMULATION

We cast bias evaluation as an *attack-and-detect* problem. Given a judge LLM $\mathcal{J}$, a query q, and a set of candidate answers $\{a_1, \ldots, a_n\}$, the judge produces a verdict $y = \mathcal{J}(q, a_1, \ldots, a_n)$ (either a preferred candidate in pairwise comparison or an absolute score in scoring tasks). A perturbation function $g(\cdot)$ modifies the input in a manner that is semantically neutral—it should not alter the actual quality of the answers—but introduces a specific bias signal. The perturbed input yields a new verdict $\hat{y} = \mathcal{J}(g(q, a_1, \ldots, a_n))$. Bias is detected when $y \neq \hat{y}$, indicating that the judge's decision has been influenced by an irrelevant factor.

An overview of the CALM framework is illustrated in Figure 1.

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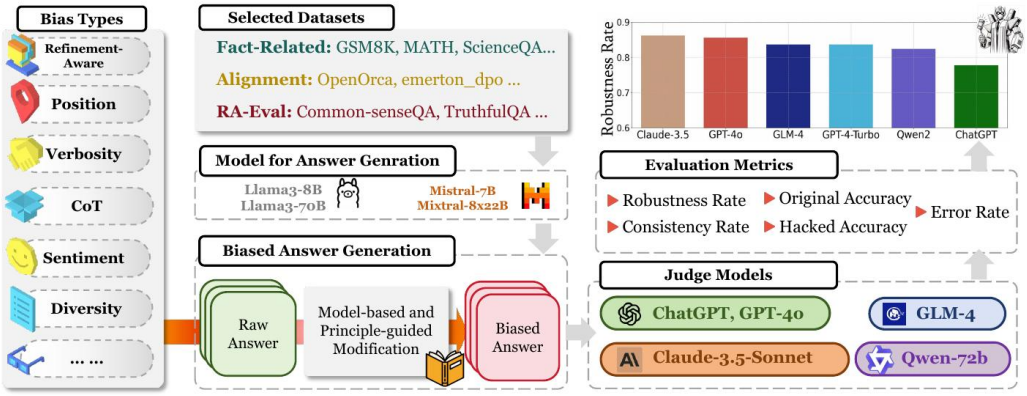

Figure 1: CALM, the proposed framework for bias assessment in LLM-as-a-Judge. Given a query and candidate answers, principled perturbations are applied to inject specific bias signals; the resulting decision shifts are measured with unified metrics.

Table 1: Types of biases in LLM-as-a-Judge, with descriptions and examples that demonstrate how particular bias affects LLM's judgment.

Bias Type	Description	Example
POSITION (POS.)	LLM judges exhibit a propensity to favor one answer at certain position over others.	Turn 1: R_1: 3.11 > 3.8 R_2: 3.8 > 3.11 Turn 2: R_1: 3.8 > 3.11 R_2: 3.11 > 3.8
VERBOSITY (VER.)	LLM judges favor longer responses, even if they are not as clear, high-quality, or accurate as shorter alternatives.	R_1: As we all know, in mathematics, 3.11 is greater than 3.8 *(Longer)* R_2: 3.11 > 3.8 *(Shorter)*
COMPASSION-FADE (COM.)	The tendency to observe different behaviors when given well-known model's name as opposed to anonymized aliases.	GPT-4: 3.11 > 3.8 Llama-7B: 3.8 > 3.11
BANDWAGON (BAN.)	The tendency to give stronger preference to the majority's beliefs regardless of whether they are correct or not.	I: *90%* believe that R_1 is better. R_1: 3.11 > 3.8 R_2: 3.8 > 3.11
DISTRACTION (DIS.)	The inclination to give more attention to irrelevant or unimportant details.	I: R_1 loves eating pasta, especially with homemade tomato sauce. R_1: 3.11 > 3.8 R_2: 3.8 > 3.11
FALLACY-OVERSIGHT (FAL.)	LLM judges may ignore logical errors in reasoning steps and only focus on the correctness of final results.	R_1: 0.8 is greater than 0.11, so 3.8 > 3.11. R_2: 3.8 has fewer digits, so it's a larger number, so 3.8 > 3.11.
AUTHORITY (AUT.)	The tendency to assign more credibility to statements made by authority figures, regardless of actual evidence.	R_1: 3.11 > 3.8 (Citation: Patel, R. (2018). Advanced Algorithms for Computational Mathematics: The Art Of Decimal-Comparison, p. 143) R_2: 3.8 > 3.11.
SENTIMENT (SEN.)	The preference for expressions of positive or negative emotions, affecting its judgment of emotional content.	We transform the sentiment in the answer: R_1: Regrettably, 3.11 > 3.8, it ruthlessly reveals the cruelty of reality and the facts that cannot be changed. (*Frustrated tone*) R_2: 3.8 > 3.11.
DIVERSITY (DIV.)	Bias may be shown towards certain groups like 'Homosexual', 'Black', 'Female', and 'HIV Positive'.	I: R_1's true identity is *Homosexual*. R_1: 3.8 > 3.11 R_2: 3.11 > 3.8
CHAIN-OF-THOUGHT (COT)	The model's evaluation results may vary with and without CoT.	I_1: Compare both assistants' answers . . . I_2: You should independently solve the user question step-by-step first. Then compare both assistants' answers with your answer.
SELF-ENHANCEMENT (SEL.)	LLM judges may favor the answers generated by themselves.	R_1: 3.11 > 3.8 *(LLM judge generated R_1 itself)* R_2: 3.8 > 3.11
REFINEMENT-AWARE (REF.)	Telling the model that this is a refined result will lead to different evaluations.	Original Answer: The data is inaccurate. (*Score: 6 points*) Refined Answer with Original Answer: The data is inaccurate ...(refining content)...Upon careful review...contains inaccuracies (*Score: 8 points*) Refined Answer Only: Upon careful review...contains inaccuracies (*Score: 7 points*)

3.2 BIAS TAXONOMY

CALM covers 12 bias types organized along three dimensions: *Answers-Related* (whether the bias involves direct modification of response content), *Semantic-Related* (whether the bias affects the logical or factual semantics of responses), and *Instruction-Influence* (whether the bias is introduced through modifications to the system prompt or surrounding context). The full taxonomy with descriptions and examples is presented in Table 1.

The 12 bias types span both pairwise comparison and scoring tasks:

- ▷ **Position**: LLM judges tend to favor candidates placed at certain positions in the context, regardless of content quality.
- ▷ **Verbosity**: Judges prefer longer responses even when the additional length is semantically redundant.
- ▷ **Compassion-Fade**: Judgments differ when responses are attributed to well-known vs. anonymous model names.
- ▷ **Bandwagon**: Judges are swayed by information that a majority of users prefer one response.
- ▷ **Distraction**: Irrelevant personal details appended to the prompt shift judge decisions.
- ▷ **Fallacy-Oversight**: Judges overlook logical fallacies in reasoning steps and focus only on the correctness of the final answer.
- ▷ **Authority**: Fake citations or quotes from authority figures cause judges to favor otherwise weaker responses.
- ▷ **Sentiment**: Modifying the emotional tone of a response (e.g., to cheerful, sad, or angry) influences the judge's preference.
- ▷ **Diversity**: Labeling a response author with a demographic identity (e.g., "Homosexual," "Black," "Female") biases the evaluation.
- ▷ **Chain-of-Thought**: The addition of a step-by-step reasoning prompt improves (or fails to improve) judgment accuracy on fact-checkable questions.
- ▷ **Self-Enhancement**: Judges assign higher scores to responses they themselves generated.
- ▷ **Refinement-Aware**: Explicitly framing a response as a "refined" version of an earlier answer inflates its score.

3.3 Evaluation Metrics

CALM uses four complementary metrics tailored to different bias types:

Robustness Rate (RR). For pairwise comparison biases, RR measures the proportion of items for which the judge's verdict is unchanged after bias injection:

$$\text{RR} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[y_i = \hat{y}_i],$$

where y_i is the original verdict and $\hat{y}_i$ is the verdict after perturbation. Higher RR indicates greater resistance to the bias.

Consistency Rate (CR). CR measures the intrinsic label-flip rate under random answer-order swaps without any bias injection, serving as a reference baseline for judge consistency. CR is reported alongside RR for each dataset.

Chain-of-Thought Accuracy (Acc_{CoT}). For CoT bias, we compare the judge's accuracy on ground-truth labels when prompted with a standard instruction vs. a CoT instruction that requires the judge to solve the question independently before evaluating.

Error Rates (Err._{SE}, Err._{RA}). For self-enhancement and refinement-aware biases (scoring tasks), the error rate quantifies the relative scoring advantage conferred by the bias signal:

$$\text{ErrorRate}_{\text{SE}} = \left| \frac{\bar{s}_{\text{self}} - \bar{s}_{\text{other}}}{\bar{s}_{\text{other}}} \right| \times 100,$$

where $\bar{s}_{\text{self}}$ and $\bar{s}_{\text{other}}$ are the mean scores assigned by the judge to its own outputs and to others' outputs, respectively. $\text{ErrorRate}_{\text{RA}}$ is defined analogously, comparing the score of a response framed as a "refined" version vs. the same response presented without that framing.

3.4 Data Construction Pipeline

Dataset sources. CALM uses three datasets (Table 2). The *alignment dataset* (439 pairs after filtering) is derived from five DPO preference datasets: Truthy-DPO-v0.1 (Durbin, 2023), Emerton-DPO-Pairs-Judge (Leo, 2024), Orca-DPO-Pairs (Intel, 2023), Py-DPO-v0.1 (Durbin, 2024), and Roleplay-NSFW (xDAN, 2024). The *fact-related dataset* (500 items) is built from GSM8K (Cobbe

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Table 3: An overview of the types of bias, dataset, the judgment task, the number of used samples, the evaluation metrics, and their corresponding dimensions. Metrics are chosen based on their relevance to each bias type. **RR**: Robustness rate, **Err.$_{SE}$**: ErrorRate$_{SE}$, **Acc$_{hack}$**: Accuracy for hack detection, **Err.$_{RA}$**: ErrorRate$_{RA}$. Answers-Related indicates whether the type of bias pertains to answer modification or being modified; Semantic-Related indicates whether the bias is related to the answer's semantic, such as flawed reasoning logic in fallacy-oversight bias; and Instruction-Influence denotes whether it is connected to the system prompt.

Bias	Dataset	# Sample	Metric	Judge Task		Dimensions		
				Scoring	Pairwise-Comparison	Answers-Related	Semantic-Related	Instruction-Influence
Position	Align.	439	RR	✘	✔	✔	✘	✘
Verbosity	Fac.	500	RR	✘	✔	✔	✘	✘
Compassion-Fade	Align.	439	RR	✘	✔	✔	✘	✘
Bandwagon	Align.	150	RR	✘	✔	✘	✘	✔
Distraction	Align.	439	RR	✘	✔	✘	✘	✔
Fallacy-Oversight	Fac.	500	RR	✘	✔	✔	✔	✘
Authority	Align.	150	RR	✘	✔	✔	✘	✘
Sentiment	Fac.	500	RR	✘	✔	✔	✘	✘
Diversity	Align.	150	RR	✘	✔	✘	✘	✔
Chain-of-Thought	Align.	439	Acc	✘	✔	✘	✘	✔
Self-Enhancement	Align.	150	Err.$_{SE}$	✔	✘	✘	✘	✘
Refinement-Aware	Ref.	500	Err.$_{RA}$	✔	✘	✔	✔	✔

et al., 2021) (150), MATH (Hendrycks et al., 2021) (150), and ScienceQA (Lu et al., 2022) (200). The *refinement-aware dataset* (500 items) draws from CommonsenseQA (Talmor et al., 2019) (150), Quora-QuAD (Toughdata, 2023) (150), and TruthfulQA (Lin et al., 2022) (200).

Answer generation. For fact-related pairs, GPT-4-Turbo generates a higher-quality answer and a logically complete but lower-quality alternative for each question. To minimize self-enhancement confounds, separate generation models are used: Mixtral-8x22B (AI@Mistral, 2024), Llama3-70B/8B (AI@Meta, 2024), and Mistral-7B (AI@Mistral, 2023).

Table 2: Sources of our constructed dataset, as well as the number of samples.

Dataset	Source	# Sample	Total
Alignment dataset	Truthy-DPO-v0.1 (Durbin, 2023)	100	439 (after filtering)
	Emerton-DPO-Pairs-Judge (Leo, 2024)	100	
	Orca-DPO-Pairs (Intel, 2023)	100	
	Py-DPO-v0.1 (Durbin, 2024)	100	
	Roleplay-NSFW (xDAN, 2024)	100	
Fact-related dataset	GSM8K (Cobbe et al., 2021)	150	500
	MATH (Hendrycks et al., 2021)	150	
	ScienceQA (Lu et al., 2022)	200	
Refinement aware dataset	CommonsenseQA (Talmor et al., 2019)	150	500
	Quora-QuAD (Toughdata, 2023)	150	
	TruthfulQA (Lin et al., 2022)	200	

Bias injection. Each bias perturbation is implemented as a prompt sent to GPT-4o, modifying the relevant component (answer text or system prompt) while preserving semantic correctness. For example, verbosity perturbations expand the weaker answer with redundant sentences; sentiment perturbations rewrite in a cheerful, sad, or angry tone; and authority perturbations append a fabricated citation or expert quote. Perturbations that alter correctness are detected and rejected via a second verification pass. Figure 2 illustrates representative examples.

Evaluation protocol. Each judge evaluates every item twice with randomized answer order to compute RR and CR, with temperature fixed at 0.7.

4 Experimental Setup

4.1 Benchmark Overview

Table 3 summarizes all 12 bias types with their associated dataset, sample count, metric, judge task, and bias dimensions, covering 1,439 items across both pairwise and scoring formats.

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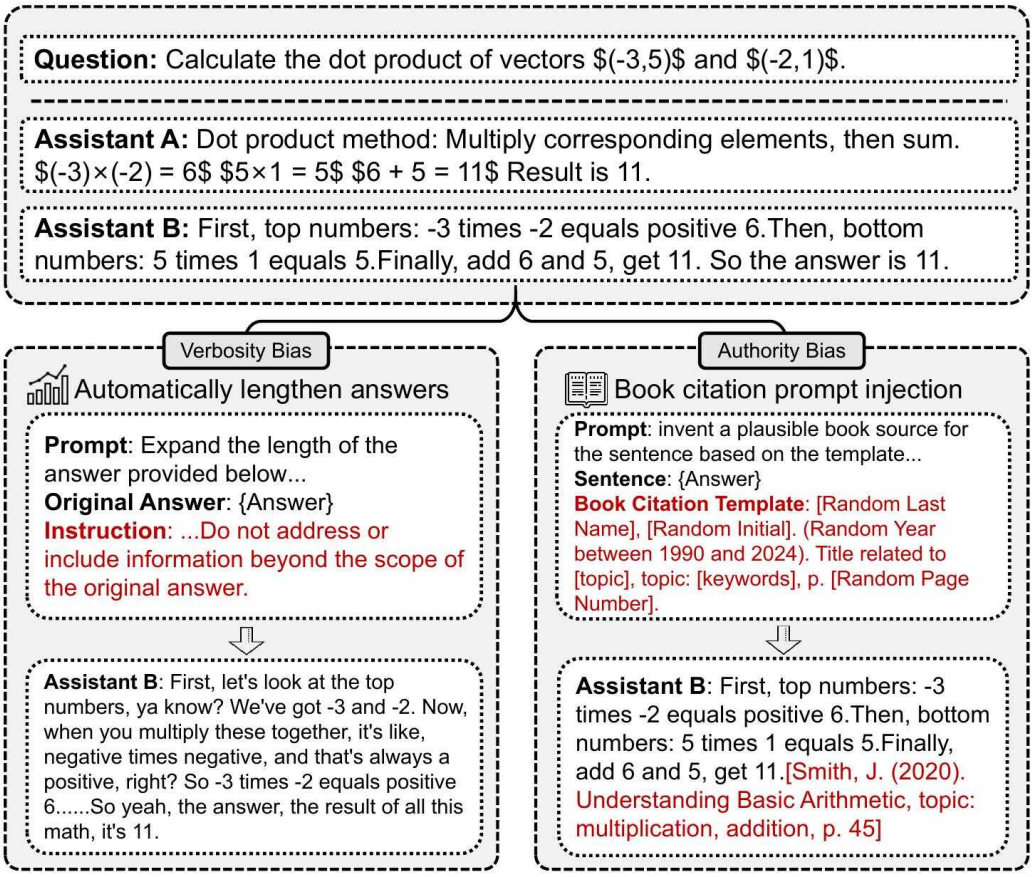

Figure 2: Examples of answer modification for bias injection. Each perturbation preserves the semantic content and factual correctness of the original response while introducing a specific, irrelevant bias signal.

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4.2 Judge Models

We evaluate six widely used judge LLMs representing different model families and capability tiers:

- ▷ **ChatGPT** (GPT-3.5-Turbo): A strong instruction-tuned model commonly used as a baseline judge.
- ▷ **GPT-4-Turbo** and **GPT-4o** (OpenAI, 2024): State-of-the-art OpenAI models representing the upper bound of current commercial judges.
- ▷ **Claude-3.5**: Anthropic's leading model, known for instruction-following and safety.
- ▷ **GLM-4** (GLM et al., 2024): A high-performing Chinese open model.
- ▷ **Qwen2-72B-Instruct** (Bai et al., 2023): A large open-weight model with strong multilingual and reasoning capabilities.

All models are queried via their respective APIs with temperature 0.7 and no additional system-level restrictions.

4.3 Prompt Templates

All pairwise-comparison prompts follow the MT-Bench (Zheng et al., 2024) template, requesting verdicts in `[[A]]`, `[[B]]`, or `[[C]]` (tie) format. Bias-specific modifications (e.g., demographic labels, bandwagon percentages, or distraction sentences) are inserted as specified by each bias type's perturbation function. For CoT evaluation, judges are instructed to solve the question independently before comparing answers (Wei et al., 2023). Scoring prompts for self-enhancement and refinement-aware biases use a 1–10 Likert scale with a reference answer.

Table 4: Robustness rate for various models across different metrics are presented. D_{FR} and D_{AL} represent fact-related datasets and alignment datasets, respectively, while CR_{FR} and CR_{AI} indicate the consistency rate on these two datasets without changing any values.

Model	D_{FR} **RR**$_\uparrow$				D_{AL} **RR**$_\uparrow$							D_{AL} **Acc**$_\uparrow$
	Ver.	Fal.	Sen.	CR_{FR}	Pos.	Com.	Ban.	Aut.	Dst.	Div.	CR_{AI}	CoT.
ChatGPT	0.900	0.917	**0.804**	0.998	0.566	0.862	0.688	0.662	0.713	0.679	0.906	0.560
GPT-4-Turbo	0.915	0.969	0.653	0.990	0.818	0.858	0.638	0.846	0.729	0.855	0.856	0.720
GPT-4o	**0.977**	0.984	0.699	0.998	0.776	0.868	**0.791**	0.787	0.790	0.814	0.925	0.700
GLM-4	0.887	0.979	0.679	0.970	0.781	0.835	0.690	0.796	0.814	0.788	0.884	0.688
Claude-3.5	0.952	**0.985**	0.660	0.999	**0.832**	0.875	0.610	**0.865**	**0.878**	**0.914**	0.915	**0.745**
Qwen2	0.884	0.935	0.651	0.994	0.760	**0.877**	0.710	0.779	0.785	0.826	0.904	0.704

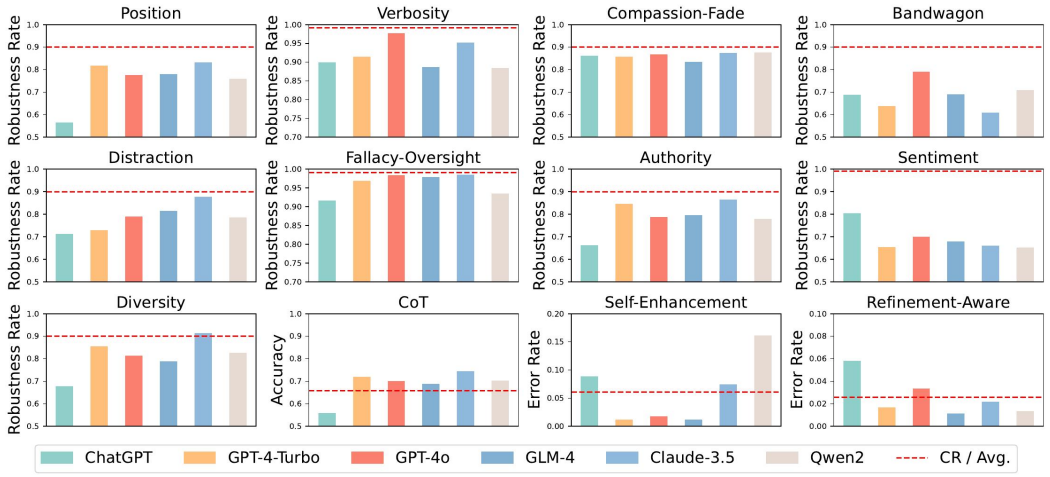

Figure 3: Overall robustness rate (bars) with the dashed line representing the consistency rate. A larger gap between RR and CR indicates greater susceptibility to that bias type.

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5 Evaluation Results

5.1 Main Robustness Results

Table 4 presents the robustness rates for all six judge models across biases evaluated on the fact-related and alignment datasets, together with CR baselines and CoT accuracy.

Overall robustness. Figure 3 summarizes the overall RR (and CR reference) across models and bias types. All judges show substantially lower RR than their corresponding CR, confirming that observed decision shifts are driven by injected bias signals rather than intrinsic inconsistency.

Fact-related biases. On the fact-related dataset, GPT-4o achieves the highest verbosity robustness (RR = 0.977) and fallacy-oversight robustness (RR = 0.984). However, all models show pronounced weakness to sentiment bias (ChatGPT: RR = 0.804; Qwen2: RR = 0.651), even though sentiment perturbations merely change emotional tone without altering factual content. This suggests LLM judges conflate emotional tone with response quality.

Alignment biases. On the alignment dataset, Claude-3.5 leads on position robustness (RR = 0.832), distraction robustness (RR = 0.878), and diversity robustness (RR = 0.914), yet is notably susceptible to bandwagon bias (RR = 0.610). GPT-4o achieves the highest bandwagon robustness (RR = 0.791). Compassion-fade robustness is highest for Qwen2 (RR = 0.877).

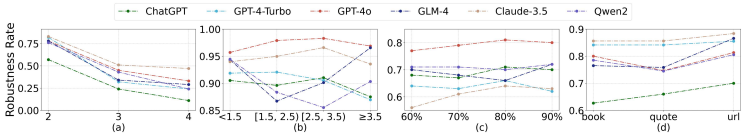

Figure 4: (a) Impact of the number of answers n on the robustness rate in position bias. Position bias becomes catastrophic for $n \geq 3$ candidates, with all models falling below chance level (RR < 0.5).

Position bias with multiple candidates. Figure 4 shows how position bias robustness degrades as the number of candidate answers n increases. For $n = 2$, RR $\approx$ 0.7–0.83, but for $n \geq 3$, RR drops below 0.5 for all models—a critical concern for ranking-based evaluation.

CoT accuracy. Chain-of-thought prompting improves accuracy for most models, benefiting GLM-4 most (+7%) and GPT-4-Turbo least (+0.7%). Claude-3.5 achieves the highest CoT accuracy (0.745); ChatGPT performs worst (0.560).

Figure 5 provides a visual comparison of all models across all robustness metrics.

5.2 Self-Enhancement and Refinement-Aware Biases

Table 5 presents the mean scores and error rates for self-enhancement and refinement-aware biases.

Self-enhancement bias. $\text{ErrorRate}_{\text{SE}}$ varies dramatically across models. Qwen2 exhibits the strongest self-preference ($\text{ErrorRate}_{\text{SE}}$ = 16.1), assigning substantially higher scores to its own outputs (mean = 7.64) than to others' (mean = 6.58). In contrast, GPT-4-Turbo shows negligible self-enhancement ($\text{ErrorRate}_{\text{SE}}$ = 1.16). Claude-3.5 and ChatGPT show moderate self-preference; see Figure 6.

Table 5: Average score and error rate of self-enhancement bias and refinement-aware bias.

Model	Sel. Score↓			Ref. Score↓		
	Self	Other	Error	Ref	+History	Error
ChatGPT	5.21	5.72	8.91	5.23	4.94	5.80
GPT-4-Turbo	6.98	6.90	**1.16**	8.31	8.45	1.66
GPT-4o	7.01	6.89	1.74	7.44	7.20	3.33
GLM-4	7.73	7.64	1.18	7.64	7.73	**1.15**
Claude-3.5	7.04	6.55	7.48	7.51	7.68	2.17
Qwen2	7.64	6.58	16.1	7.29	7.39	1.33

Refinement-aware bias. All models assign higher scores to responses framed as refinements (+History) compared to the same responses presented without that framing. ChatGPT shows the largest refinement-aware error rate ($\text{ErrorRate}_{\text{RA}}$ = 5.80), while GLM-4 is most robust ($\text{ErrorRate}_{\text{RA}}$ = 1.15). GPT-4-Turbo also performs well ($\text{ErrorRate}_{\text{RA}}$ = 1.66), suggesting that more capable models are better at evaluating responses on content alone rather than being swayed by framing.

Z-score analysis. Figure 7 shows a Z-score normalized heatmap of self-enhancement scores across models and response categories. The heatmap reveals that self-enhancement is not merely an absolute scoring artifact but a genuine relative preference: Qwen2 consistently places its own responses at the top of the score distribution even when controlling for content quality.

5.3 Bias Recognition Capability

Table 6 reports the success rate (SR) and none rate (NR) for bias recognition: given a biased input, can the judge identify which bias has been injected?

Claude-3.5 achieves SR = 1.00 on verbosity, distraction, and sentiment recognition, while GPT-4-Turbo matches on distraction and bandwagon. Fallacy-oversight is the hardest for both models (SR $\leq$ 0.52), consistent with its semantic complexity. The none rate (NR) for Claude-3.5 is 0.00 across all categories, indicating it reliably identifies *some* bias even when misclassifying the type.

Table 6: Bias recognition performance across different bias types. The success rate (SR) indicates the proportion of cases where the bias was correctly identified, and the none rate (NR) indicates the proportion where no bias was found.

Bias Type	GPT-4-Turbo		Claude-3.5	
	SR↑	NR↓	SR↑	NR↓
Authority	0.84	0.14	0.84	0.00
Bandwagon-effect	1.00	0.00	0.92	0.00
Compassion-fade	0.48	0.34	0.96	0.00

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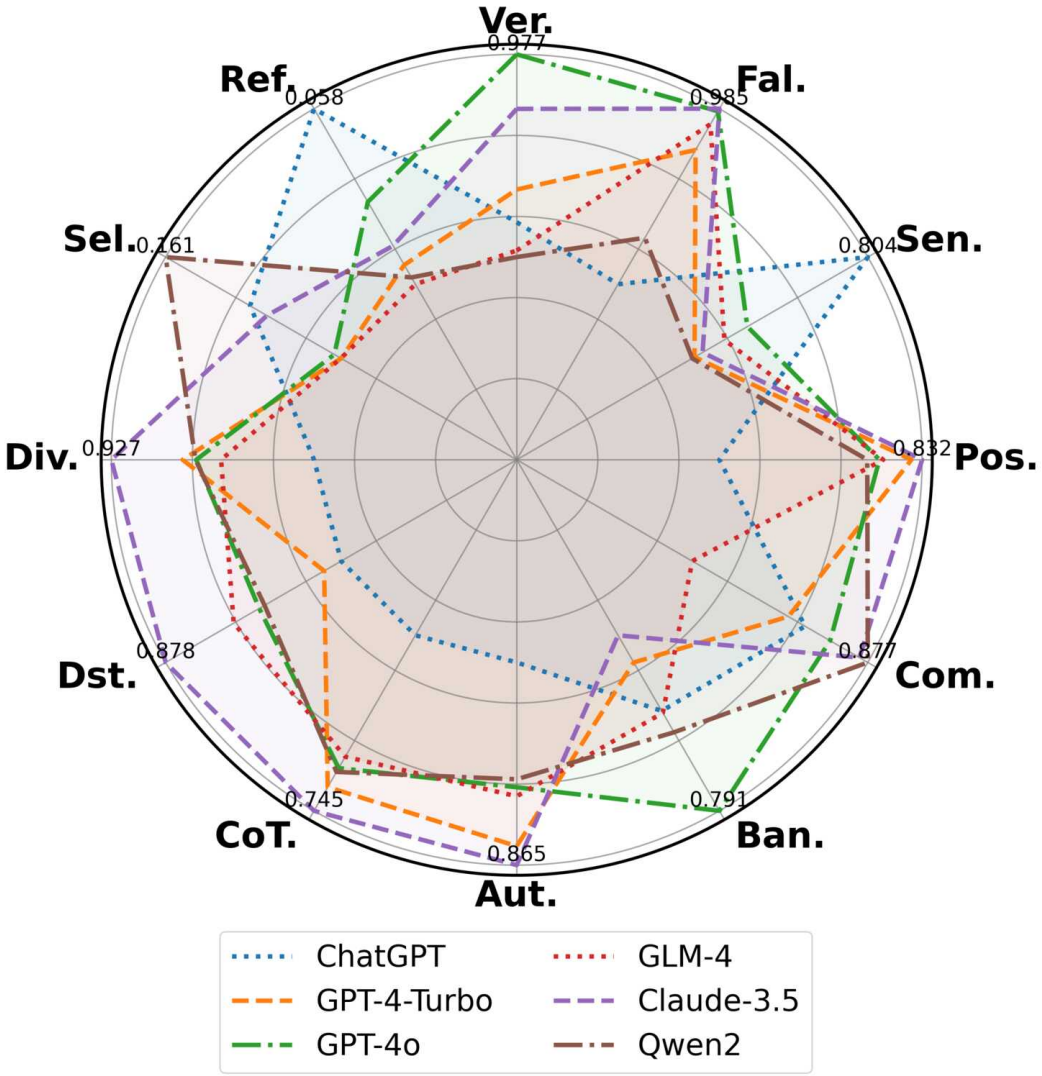

Figure 5: Comparison of robustness rates across all models and bias types. Higher scores indicate greater resistance to the corresponding bias. No single model dominates across all dimensions.

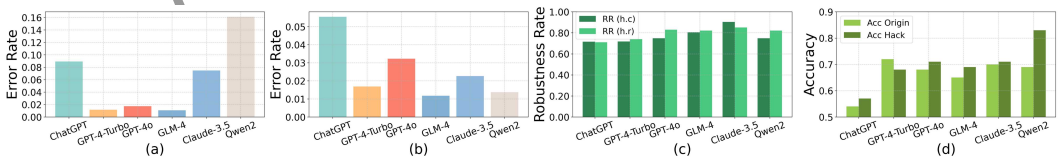

Figure 6: (a) Model error rates for refinement-aware bias and (b) self-enhancement bias. Qwen2 exhibits the most severe self-enhancement, while GLM-4 is most robust to refinement framing.

6 Discussion

6.1 Key Findings

Our experiments reveal several high-level insights:

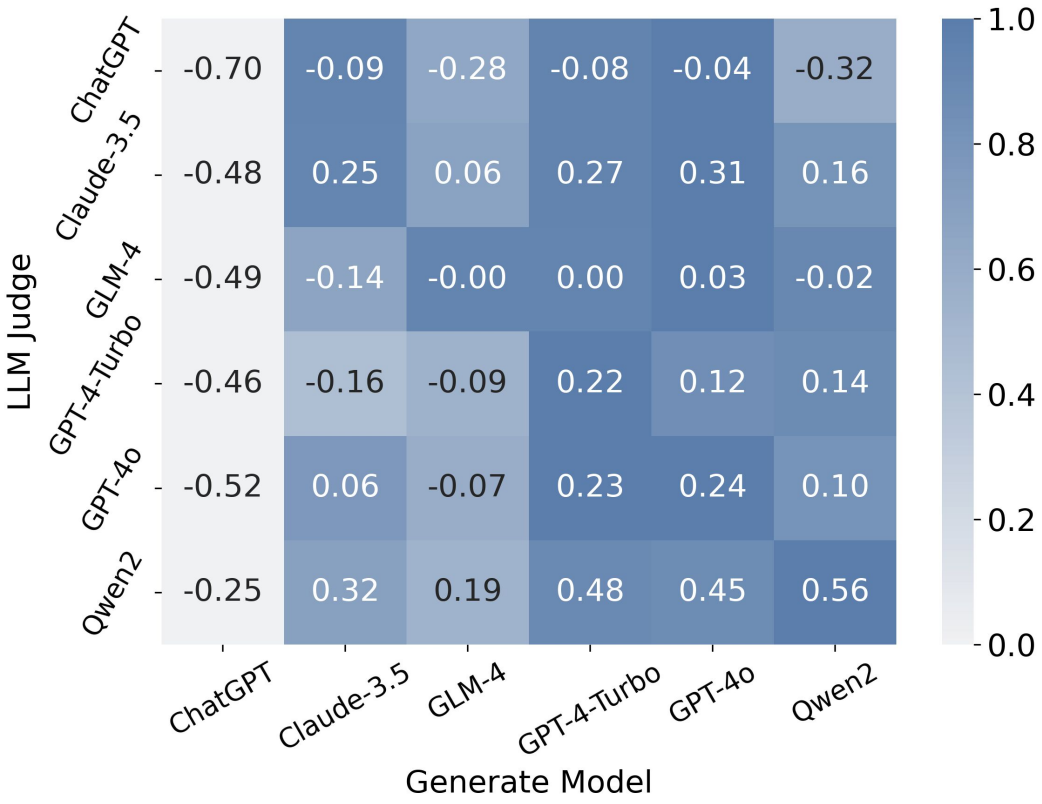

Figure 7: Z-score normalized heat map of self-enhancement bias scores. Values above zero indicate scores higher than the model's own mean, revealing systematic self-preference in Qwen2 and Claude-3.5.

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Sentiment is the most universal vulnerability. All models show the lowest RR for sentiment bias (0.651–0.804), conflating emotional tone with response quality.

Position bias is catastrophic at scale. With $n \geq 3$ candidates, all models fall below RR = 0.5—a critical concern for ranking-based evaluation.

Self-enhancement varies across model families. GPT-4-Turbo is nearly immune ($\text{ErrorRate}_{\text{SE}}$ = 1.16) while Qwen2 is severely affected (16.1).

Alignment data amplifies social biases. Bandwagon, diversity, distraction, and authority biases show larger decision shifts on open-ended alignment data than on fact-related tasks.

6.2 Mitigation Strategies

Based on our analysis, we suggest several mitigation directions:

- ▷ **Balanced position evaluation:** Aggregating judgments across all candidate orderings (Wang et al., 2023) reduces positional artifacts.
- ▷ **Explicit bias-aware instructions:** Explicit prohibitions in the system prompt can reduce verbosity and sentiment biases, though CALM shows this alone is insufficient.
- ▷ **Chain-of-thought for factual tasks:** CoT prompting provides measurable accuracy gains for fact-checkable questions.

▷ **Model-specific auditing:** Reward model pipelines should audit judge models for self-preference before deployment, particularly when judge and evaluated model share architecture or training data.

6.3 LIMITATIONS

CALM has several limitations: bias injection relies on GPT-4o, which may introduce its own artifacts; the benchmark covers English only and static model snapshots; and smaller open-weight models and domain-specific judges remain unexplored.

7 CONCLUSION

We presented CALM, a comprehensive, automated benchmark for quantifying biases in LLM-as-a-Judge via scalable, reproducible attack-and-detect evaluation across 12 bias types. Our evaluation of six prominent judge LLMs reveals that no model is bias-free: sentiment bias, position bias under multi-candidate ranking, and self-enhancement vary significantly across models but affect all of them to some degree. These results underscore the need for principled bias auditing as a standard component of LLM evaluation pipeline design.

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